

HARP-NET CKM: Prior Anchored Deep Learning for Dense UAV Channel Knowledge Maps

Guillem Moreno Garcia, Evgenii Vinogradov, and Sergi Abadal

Abstract—Channel knowledge maps (CKMs) can turn expensive ray tracing into fast radio planning, but dense UAV CKM generation remains difficult when channel attenuation, delay spread, and angular spread must transfer to unseen cities. This paper presents HARP-NET CKM (Height Aware Residual Prior Network for Channel Knowledge Map prediction), a prior anchored surrogate. The method first calibrates frozen physical and statistical priors on training cities only: coherent two-ray and NLoS attenuation priors, plus log domain DS/AS spread priors. A shared height conditioned residual network then predicts bounded, distribution aware corrections instead of full maps from scratch. The model receives a building height map, support/visibility masks, five frozen prior maps, and scalar UAV transmitter height, and outputs three dense 513×513 maps. On a 14 city held out test split it reaches 1.6947 dB attenuation RMSE, 26.6882 ns delay spread RMSE, and 11.5253° angular spread RMSE, improving the frozen priors for all three targets. The result supports a practical design rule: keep stable radial, visibility, height, and spread structure in auditable priors, and spend neural capacity on the residual structure that remains.

Index Terms—channel knowledge maps, UAV communications, radio map prediction, channel attenuation, delay spread, angular spread, Gaussian mixture models, physics informed learning.

I. INTRODUCTION AND RELATED WORK

Channel knowledge maps (CKMs) store or predict radio channel quantities as functions of position and environment, enabling faster planning and adaptation than repeated ray tracing or exhaustive measurements [1], [2]. In UAV enabled 6G networks, CKMs are especially attractive because altitude changes both the geometry and the line of sight probability of the link [3], [4]. A practical CKM surrogate, however, must generalize to urban layouts not seen during training and must not hide hard propagation regimes behind a single global score.

The central difficulty is that the three targets do not behave like one smooth image regression problem. LoS attenuation has strong radial, height dependent and two-ray structure; NLoS attenuation depends on blocked link regimes; and delay and angular spread combine near zero LoS values with heavier NLoS tails. Development experiments confirmed that a generic dense regressor can draw plausible maps while still hiding NLoS mode collapse or poor spread tails. The final design therefore separates the task into train only priors for stable structure and a bounded neural residual for what remains.

The contribution is HARP-NET CKM, the Height Aware Residual Prior Network for Channel Knowledge Map prediction. It is a dense CKM recipe for one fixed UAV evaluation

setting: strict city holdout, valid receiver masking, explicit LoS/NLoS and height reporting, priors calibrated only on the training split, and a neural correction around those priors. The paper also reports city, height and visibility diagnostics so the final average is not presented as a single opaque score.

A. Radio Map and Channel Parameter Predictors

Dense radio map prediction is often framed as image to image regression. RadioUNet [5], PMNet, and the ICASSP path loss challenge family showed that U-Net style models can be strong dense channel attenuation baselines [6]–[8]. Later work adds attention, equivariance, domain shift handling, corridor weighting, or digital twin generation [9]–[13]. These papers give the closest dense channel attenuation context, but not the same setting: continuous UAV transmitter height, strict city holdout on CKM samples, and three dense targets (channel attenuation/PL, DS, AS).

CKM work also emphasizes that a radio map is not merely an image but a position indexed channel side information object [1], [2], [14]. A random image split can place visually similar urban layouts in both training and testing; a city split tests transfer to unseen layouts, so the evaluation protocol is part of the method.

Physics informed and hybrid methods address a different weakness of pure image regression: the model should not need to rediscover stable propagation effects from pixels alone. ReVeal [15] uses physics informed constraints for radio environment mapping, while classical air to ground, COST/Hata, and two-ray models encode known distance, height, reflection, and elevation angle structure [3], [16]–[21]. This paper uses those models only as shape priors; all deployment coefficients are calibrated from training cities under the same CKM masking convention used at inference.

The spread side has fewer direct dense map references. Yang et al. predict A2G delay spread with classical regressors [22]; Huang et al. use a Transformer style UAV mmWave predictor for received power, RMS delay spread, and RMS angular spread [23]; and Mi et al. predict mmWave channel attenuation (PL), DS, AS, and K -factor from measurement linked point clouds [24]. A closer map based reference is the multitask super resolution model of Wang et al. [25], which reconstructs path loss, delay spread, RMS azimuth/elevation angular spread and LoS/NLoS maps from lower resolution channel maps. This is still a different input contract: the low resolution channel field already contains sampled DS/AS structure, whereas this paper generates dense 513×513 CKM maps from topology, support masks, priors and continuous UAV height for unseen cities.

Finally, heteroscedastic regression [26], mixture density wireless predictors [27]–[29], and stochastic radio map models [30] all point to the same issue: NLoS and spread targets are not well represented by a single symmetric residual. For DS/AS, the closest external references are therefore context rather than direct CKM benchmarks.

II. DATASET AND EVALUATION SETTING

Each sample is a 513×513 urban topology map from the ray traced CKM setting used in this work. The carrier is fixed at 7.125 GHz, so frequency is not a learned variable; the main transmitter degree of freedom is height. There is one UAV transmitter at the map centre. The transmitter horizontal position is fixed, but its antenna height h_{tx} varies by sample and is treated as a central conditioning variable rather than as metadata. The released CKM files call this node a UAV, but the model setting is an elevated 7.125 GHz transmitter within the sampled height range, so the same setup can also represent other elevated radio access nodes under the same simulation assumptions. The spatial resolution is $1 \text{ m} \times 1 \text{ m}$ per pixel. Every non building pixel is treated as an active receiver location, with the receiver antenna placed at $h_{rx} = 1.5 \text{ m}$ above the ground. Building pixels are excluded from the receiver set rather than treated as easy zero error targets. The model predicts three aligned dense maps:

$$y_t \in \{\text{PL (dB)}, \text{DS (ns)}, \text{AS (deg)}\}.$$

The CKM HDF5 field and implementation call the first target `path_loss` and retain the abbreviation PL. In this paper, PL is interpreted as the dense channel attenuation map in dB, while *path loss* is kept for standard propagation formulas, cited path loss models, and dataset/code names that use that term. Let $m_i(p) = 1$ if pixel p of sample i is one of these valid ground receivers, and zero otherwise. For a split $\mathcal{S}$, the target specific pixel weighted RMSE is

$$\text{RMSE}_t = \sqrt{\frac{\sum_{i \in \mathcal{S}} \sum_p m_i(p) (\hat{y}_{i,t}(p) - y_{i,t}(p))^2}{\sum_{i \in \mathcal{S}} \sum_p m_i(p)}}. \quad (1)$$

The same receiver-mask formula is used for all three targets. For DS and AS, each valid ground pixel is still a receiver with a ray-traced ground truth spread value, so the reported spread RMSE is the dense prediction compared with the dense CKM target over all valid ground receivers. To check whether the residual model improves spatial structure rather than only global scale, this paper also reports map correlation:

$$\text{MapCorr}_t = \frac{\sum_{i \in \mathcal{C}_t} N_i \rho_{i,t}}{\sum_{i \in \mathcal{C}_t} N_i},$$

$$\rho_{i,t} = \frac{\tilde{\mathbf{y}}_{i,t}^\top \tilde{\mathbf{y}}_{i,t}}{\|\tilde{\mathbf{y}}_{i,t}\|_2 \|\tilde{\mathbf{y}}_{i,t}\|_2}.$$

where $\Omega_i = \{p : m_i(p) = 1\}$, $N_i = |\Omega_i|$, and $\tilde{\mathbf{y}}_{i,t}$ and $\hat{\mathbf{y}}_{i,t}$ are the valid pixel vectors after subtracting their own per-map means. The set $\mathcal{C}_t$ excludes maps with fewer than two valid pixels or zero variance in either vector. This is Pearson correlation after each map is centred and normalized over its own valid receiver pixels, then averaged by valid pixel count.

TABLE I
EXPERIMENTAL SETTING USED FOR ALL REPORTED FINAL NUMBERS.

Item	Setting
Map resolution	513×513 , $1 \text{ m} \times 1 \text{ m}$ pixels
Carrier / receiver	7.125 GHz; $h_{rx} = 1.5 \text{ m}$
Transmitter	One centred UAV; stored $h_{tx} \approx 12 \text{ m}$ to 478 m , nominal beta law 10 m to 500 m
Inputs	Topology, ground/visibility masks, frozen priors, scalar height code
Targets	Channel attenuation / PL (dB), DS (ns), AS (deg), one dense map per target
Stored target ranges	PL 65 dB to 184 dB, DS 0 ns to 910 ns, AS 0° – 180° over all valid receivers
Metric mask	Non building ground receivers only; building pixels excluded
Split policy	City holdout; validation selects the checkpoint, test is final only
Samples	Train 10840; validation 2750; test 2590; all 16180

Higher values mean that high and low regions are placed more similarly to the ray traced target; RMSE remains the primary unit aware metric.

The split is city based rather than random over pixels or tiles. This matters: random splits would allow the model to see nearly identical urban morphology during training and testing. The final held out test split contains 2590 samples from cities not used to optimize model weights. Validation is used for model selection; test is used only for the final generalization claim. Table II lists the exact city assignment. Unless explicitly stated otherwise, every final model result reported below refers to the separate 2590-sample, 14-city test split.

The transmitter height distribution is intentionally not uniform. Direct comparison against the stored HDF5 histogram shows that the closest tested affine version of the approximate double beta sampler is the original $h_{\text{design}} \approx 10 + 490X$, with

$$X \sim 0.75 \text{Beta}(3, 23) + 0.25 \text{Beta}(4, 4),$$

and nominal support 10 m to 500 m . Reported results use the empirical HDF5 realization: h_{tx} spans 11.61 m to 477.54 m , with mean 114.02 m and median 75.66 m . Heights above 478 m are possible under the nominal law but rare, so the observed maximum is consistent with finite sampling. Let $\Delta h = 3 \text{ m}$, $B_k = [10 + 3k, 13 + 3k)$, and h_i be the stored height of sample i . The reporting histogram is

$$\hat{f}_{\Delta h}(h) = d_k, \quad h \in B_k,$$

$$d_k = \frac{n_k}{N \Delta h}, \quad n_k = \sum_{i=1}^N \mathbf{1}\{h_i \in B_k\}, \quad N = 16180. \quad (2)$$

For readability in Fig. 1, the plotted curve linearly interpolates between the bin centre values (c_k, d_k) , with $c_k = 11.5 + 3k$. All database counts are 3972 samples below 50 m , 8456 from 50 m to 150 m , 2490 from 150 m to 300 m , and 1262 above 300 m ; the corresponding training counts are 2676, 5654, 1644, and 866. This skew also motivates the clipped logarithmic height feature used inside the frozen priors: its denominator $\log(401)$ corresponds to a practical 400 m transmitter to receiver height difference. The HDF5 maximum is higher, but the higher altitude tail is thin after city and regime splitting, so the normalization is chosen as a robust saturation scale rather than as the absolute sample maximum.

TABLE II
CITY HOLDOUT ASSIGNMENT USED BY THE DISTRIBUTION FIRST AND FINAL NEURAL MODELS.

Split	Samples	Cities	City names
Train	10840	37	Abidjan, Abu Dhabi, Alexandria, Antigua Guatemala, Athens, Bangalore, Bangkok, Bath, Berlin, Bratislava, Brugge, Buenos Aires, Canberra, Chefchaouen, Dalian, Hanoi, Ho Chi Minh City, Hyderabad, Kinshasa, Leuven, Ljubljana, Luanda, Lyon, Manaus, Minsk, Nagpur, Paris, Rio de Janeiro, Santiago, Shanghai, Siem Reap, St. Louis, Thane, Toledo, Valparaiso, Venice, Vienna.
Validation	2750	15	Bilbao, Chiang Mai, Cleveland, Dhaka, Dubrovnik, Fortaleza, Marrakesh, Mexico City, Munich, Nazareth, Nice, Salvador, Segovia, Tunis, Victoria.
Test	2590	14	Barcelona, Beijing, Carcassonne, Copenhagen, Halifax, Jaipur, Johannesburg, Key West, Kuala Lumpur, Osaka, Phuket, Pune, Surat, Vancouver.

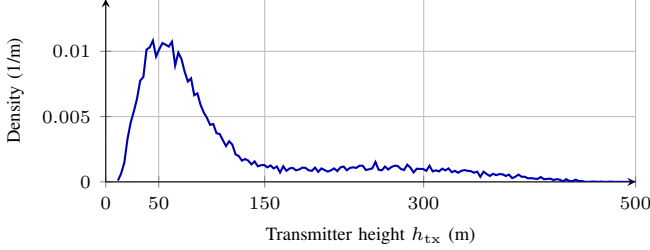


Fig. 1. Empirical transmitter height density read directly from the CKM HDF5 database using the 3 m histogram of (2), linearly interpolated between bin centres for display. Most training support lies in the low and middle altitude bands.

A. Quantized Targets and Continuous Outputs

The released CKM targets are not ideal real valued measurements. Channel attenuation is stored on an approximately 1 dB grid, while delay spread and angular spread are strongly integer valued and visibly staircase like in many maps. The model outputs are not quantized to these grids: the residual network emits continuous values, and all reported metrics compare those continuous outputs against the stored targets. This matters for interpretation: the final path loss MAE is 0.9773 dB, close to the label step, while RMSE remains 1.6947 dB because it emphasizes hard tails. Less quantized labels would likely expose more of the continuous surrogate’s resolution, especially for small residual improvements and spread map boundaries.

III. HARP-NET CKM FRAMEWORK

A. Why the Final Method Is Hybrid

HARP-NET CKM separates stable structure from the residual structure that remains hard to model analytically. First, the attenuation prior captures LoS distance, height and two-ray effects and adds a calibrated NLoS branch for blocked regions. Second, the spread priors use log domain regime calibration for delay spread and angular spread. Third, the neural model receives these frozen maps and predicts only bounded residual corrections. This division keeps the predictable part of the map auditable and reserves learning capacity for the local errors that the frozen estimators cannot explain.

In this paper, a *prior* means a frozen estimator fitted only on training cities, not a hand tuned curve evaluated on the test set. The prior sees the same support information available at deployment: centred transmitter geometry, transmitter height, topology derived morphology, and a LoS/NLoS mask. In the CKM benchmark runs, the stored HDF5 mask is used so every reported number uses the same simulator support labels. For new datasets where only topology and height are available, the generator ray casts the LoS/NLoS mask from those two

TABLE III
INPUT/OUTPUT DEFINITION OF THE FULL FRAMEWORK AND FINAL NEURAL MODEL.

Block	Variables
External framework inputs	Building height map and UAV transmitter height h_{tx} , stored as <code>uav_height</code> in the CKM files.
Computed support maps	LoS/NLoS visibility mask and valid ground receiver mask.
Frozen prior maps	Combined path loss prior, LoS path loss prior, NLoS path loss prior, delay spread prior, and angular spread prior.
Inputs to HARP-NET CKM	Topology map, transmitter height conditioning, ground mask, LoS mask, NLoS mask, and the five frozen prior maps.
Final outputs	Channel attenuation, delay spread, and angular spread maps.

inputs; these generated masks are the operational support masks consumed by the priors and by HARP-NET CKM. Once the prior coefficients and fallback tables are stored, no gradient update is applied to them inside the neural training loop. This separation is important because it lets the final residual model be judged against a demanding baseline: the model must improve an already calibrated predictor rather than merely beat FSPL.

The two lightweight regressors used inside the priors are standard least squares tools. Ordinary least squares (OLS) fits linear coefficients by minimizing training city squared error without regularization. Ridge regression uses the same linear design matrix but adds an ℓ_2 penalty, which stabilizes sparse height, topology, and visibility regimes.

Thus the full generator has two external inputs and three final outputs. The neural residual model has nine spatial map channels plus a separate scalar transmitter height conditioning path because the framework has already computed or loaded the visibility mask, receiver mask, and frozen prior maps.

B. Frozen Priors

The path loss prior combines a coherent two-ray LoS branch with a calibrated NLoS branch. More precisely, (4) is the Friis/free space dB term and (5) is the standard coherent direct plus ground reflected image source construction [16], [17], also used in low altitude A2G two-ray modeling [21]. The fitted ρ , ϕ , β_{LoS} , and r curves are not taken from those references; they are CKM training split calibration terms wrapped around that two-ray structure. In particular, $\rho(h_{tx})$ is an effective reflected field amplitude and $\phi(h_{tx})$ is the fitted phase offset of that reflected field relative to the direct ray; neither quantity is a material Fresnel coefficient. The NLoS branch uses the COST231/Hata urban macro expression as a coarse blocked link trend [18], combines it with an Al-Hourani style elevation dependent A2G excess loss envelope [19], and keeps the LoS/NLoS regime separation used in UAV

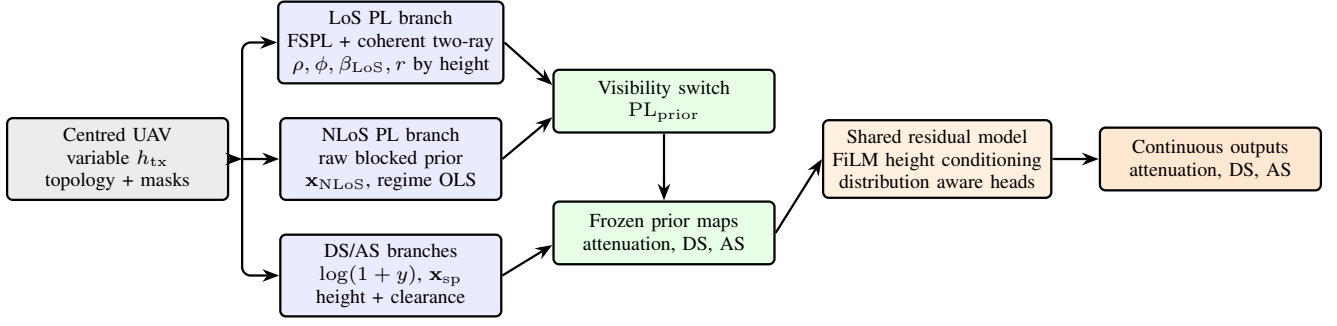


Fig. 2. Compact HARP-NET CKM pipeline. The transmitter is always centred, but its height varies by sample. Frozen height aware priors are produced first and then used as channels for the final HARP-NET CKM model.

and standardized channel model literature [3], [20]. The spread priors use the log domain treatment of large scale delay/angle spread parameters from 3GPP/WINNER style models [20], [31]; the UAV empirical literature motivates the elevation and angular spread terms rather than supplying the CKM coefficients directly [32], [33]. These references define the shape assumptions; all numerical coefficients used here are still fitted on the training cities only. The UAV remains horizontally centred, so d_{2D} is the per pixel distance from the centre transmitter to each receiver. The height h_{tx} , however, changes from sample to sample and is one of the main physical degrees of freedom in the method. For receiver height h_{rx} and carrier frequency f_{MHz} , the direct and reflected distances are

$$\begin{aligned} d_{LoS} &= \sqrt{d_{2D}^2 + (h_{tx} - h_{rx})^2}, \\ d_{ref} &= \sqrt{d_{2D}^2 + (h_{tx} + h_{rx})^2}. \end{aligned} \quad (3)$$

For the positive UAV and receiver heights used here, d_{ref} is strictly larger than d_{LoS} ; equality would occur only in the limiting geometric case where one antenna lies exactly on the ground plane. The LoS branch starts from

$$FSPL = 32.45 + 20 \log_{10} \left(\frac{d_{LoS}}{1000} \right) + 20 \log_{10}(f_{MHz}) \quad (4)$$

and adds an effective coherent reflection correction

$$C_{2ray} = -20 \log_{10} \left| 1 + \rho(h_{tx}) \frac{d_{LoS}}{d_{ref}} e^{-j(k(d_{ref} - d_{LoS}) + \phi(h_{tx}))} \right|. \quad (5)$$

The final LoS prior is

$$PL_{LoS} = FSPL + C_{2ray} + \beta_{LoS}(h_{tx}) + r(d_{2D}, h_{tx}), \quad (6)$$

where β_{LoS} is a height dependent bias and r is a smoothed radial residual fit. The direct distance d_{LoS} sets the free space baseline, d_{ref} supplies the image source ground reflected path, and $k(d_{ref} - d_{LoS})$ is the reflected wave phase lag. The calibrated $\rho(h_{tx})$, $\phi(h_{tx})$, $\beta_{LoS}(h_{tx})$, and $r(d_{2D}, h_{tx})$ curves are fitted in 5 m height bins, smoothed, interpolated at inference, and guarded to the [20, 185] dB range. The notation is local: $\rho(h_{tx})$ is the LoS reflection amplitude, whereas δ_k denotes local building density; similarly $\beta_{LoS}(h_{tx})$ is a LoS bias, while b_q^{nat} , b_{top} , and b_{41} belong to the spread prior.

The LoS calibration is deliberately structured rather than fully flexible. It is fitted only on valid LoS ground pixels from training cities. First, the effective reflection amplitude ρ and phase ϕ are fitted in 5 m transmitter height bins to reproduce

the ring like constructive and destructive interference visible in LoS channel attenuation maps. Second, the per height bias b removes a remaining vertical offset after the coherent two-ray fit. Third, the radial residual table $r(d_{2D}, h_{tx})$ averages the remaining error by distance from the centred transmitter, smooths it across radius, clips it to a small dB range, and stores it by height. At inference, the actual h_{tx} selects/interpolates these calibration curves, so the LoS prior changes continuously with antenna height even though the UAV horizontal coordinate remains fixed at the map centre.

This explicit LoS branch is diagnostic: many CKM LoS errors are radial rings around the centred transmitter rather than building edge artifacts. The two-ray plus radial table removes that height/radius component before the network spends capacity on local visibility and morphology corrections.

The raw NLoS propagation prior is the maximum of two coarse branches: a COST231 shaped urban loss branch and an A2G NLoS elevation envelope. These are subterms of the raw blocked prior, not separate final models. The first term below is the COST231/Hata urban macro formula adapted as a shape prior for the CKM geometry, not as an assertion that the original model is directly valid at every transmitter height:

$$\begin{aligned} PL_C &= 46.3 + 33.9 \log_{10}(f_{MHz}) \\ &\quad - 13.82 \log_{10}(h_{tx}) - a(h_{rx}) \\ &\quad + \eta(h_{tx}) \log_{10}(d_{km}) + 3, \\ \eta(h_{tx}) &= 44.9 - 6.55 \log_{10}(h_{tx}), \end{aligned} \quad (7)$$

where C denotes the COST231 shaped branch, $d_{km} = \max(d_{2D}/1000, 0.001)$, and $a(h_{rx}) = (1.1 \log_{10} f_{MHz} - 0.7)h_{rx} - (1.56 \log_{10} f_{MHz} - 0.8)$ is the small/medium city mobile height correction. The +3 dB term is retained as the COST231 metropolitan correction inside this empirical basis; it is not a claim that the original COST231 deployment domain transfers unchanged to 7.125 GHz UAV links. The second term is a conservative A2G NLoS envelope: it keeps altitude/elevation structure from coverage models, then lets the calibration fitted on training cities correct the fixed excess loss constants:

$$\begin{aligned} PL_{A2G, NLoS} &= \lambda_0(h_{tx}, f) + \beta_{NLoS} \\ &\quad + A_{NLoS} \exp \left(-\frac{90 - \theta^\circ}{\tau_{NLoS}} \right), \end{aligned} \quad (8)$$

with $\lambda_0 = 20 \log_{10}(4\pi h_{tx} f/c)$, $\beta_{NLoS} = -16.16$ dB, $A_{NLoS} = 12.0436$, and $\tau_{NLoS} = 7.52^\circ$. The calibration fitted

on training cities then bends these fixed constants to the CKM ray traced setting. The raw blocked link prior is the conservative envelope

$$P_0 = \text{PL}_{\text{NLoS,raw}} = \max(\text{PL}_C, \text{PL}_{\text{A2G,NLoS}}). \quad (9)$$

The maximum is deliberately pessimistic: whichever raw branch predicts stronger attenuation is kept. The retained raw attenuation is then corrected by a per regime linear calibration fitted on training cities:

$$\text{PL}_{\text{NLoS}} = \text{clip}(\mathbf{x}_{\text{NLoS}}^\top \mathbf{w}_{\text{NLoS},r}, 20, 185). \quad (10)$$

The path loss prior is therefore not one formula with a small correction. It is two different calibrated feature extractors. The LoS side calibrates height binned two-ray amplitude/phase, height bias, and radial residual curves. The NLoS side adds calibration terms for height, building topology, and density. This feature choice follows the same modelling instinct as COST 231 Walfisch Ikegami: NLoS urban loss should depend on the surrounding built form, not only on range and frequency [18]. The Walfisch Bertoni component treats rows of buildings as repeated diffraction screens [34]. The Ikegami street level component motivates descriptors such as building height, street width, street orientation, and mobile height [35]. In the CKM maps these coarse planning variables are replaced by directly observed local windows: building density, mean building height, and NLoS support fractions. A modern machine learning analogue is Juang's path profile work, where building profile statistics are used in a linear model to explain urban path loss predictions [36]. The raw blocked link prior is therefore only one feature in a 15 term morphology vector:

$$\begin{aligned} \mathbf{x}_{\text{NLoS}} = [& P_0^2, P_0, \log(1 + d_{2D}), \delta_{15}, \delta_{41}, \\ & h_{15}, h_{41}, \delta_{41} \log(1 + d_{2D}), n_{15}, \\ & n_{41}, n_{41} \log(1 + d_{2D}), \sigma_{\text{shadow}}, \theta_{\text{norm}}, \\ & n_{41} \theta_{\text{norm}}, 1]^\top, \end{aligned} \quad (11)$$

where P_0 is the raw NLoS prior, δ_{15} and δ_{41} are the fractions of building pixels in 15×15 and 41×41 windows, h_{15} and h_{41} are local building height summaries, n_{15} and n_{41} summarize local NLoS support, σ_{shadow} is an elevation dependent shadowing proxy, and θ_{norm} is the normalized elevation angle feature. The interaction $\delta_{41} \log(1 + d_{2D})$ lets density matter differently with range, $n_{41} \log(1 + d_{2D})$ lets local blockage support vary with range, and $n_{41} \theta_{\text{norm}}$ activates when obstruction and elevation jointly matter. The constant 1 is a per regime bias term. The deployed path loss prior is selected by the geometric visibility mask:

$$\text{PL}_{\text{prior}} = m_{\text{LoS}} \text{PL}_{\text{LoS}} + (1 - m_{\text{LoS}}) \text{PL}_{\text{NLoS}}. \quad (12)$$

The regime index r is not a learned latent class. It is selected from observable scene descriptors: topology/morphology class, LoS/NLoS state, and antenna height group, with fallback calibrations when an exact regime has too few training pixels. Thus the prior is auditable: one can inspect the raw blocked estimate P_0 , the local morphology terms, the selected regime, and the final linear correction separately.

This NLoS branch is intentionally less physical than the LoS branch. The visibility mask says that the direct ray is blocked,

but it does not say which street canyon, rooftop edge, or local clutter feature dominates the strongest remaining path. The calibrated feature vector therefore treats COST/A2G values as one input among morphology summaries. It is a compromise between a fully analytic blocked link law, which was too rigid, and a deep model, which was less transparent when trained from scratch.

Table IV makes this provenance explicit. The only pieces taken directly as literature structures are the Friis/two-ray geometry, the COST231/Hata urban trend, the A2G elevation loss shape, and the log domain spread convention. The reflection curves, radial residuals, OLS/ridge weights, fallback hierarchy, and morphology windows are fitted or computed inside this CKM dataset pipeline.

The LoS two-ray parameters are fitted by a small grid search rather than by a neural optimizer. For each 5 m height bin B_h , candidate values $\hat{\rho} \in \{0, 0.05, \dots, 1.5\}$ and $\hat{\phi} \in [-\pi, \pi]$ on a 48-point phase grid are evaluated on valid LoS training pixels. The upper end of this grid is a robustness allowance for an effective CKM fitted coefficient, not a claim that a passive Fresnel reflection coefficient exceeds unity; the smoothed deployed reflection amplitude curve stored by the final prior remains below unity. For each candidate pair, the dB bias has the closed form value

$$\hat{\beta}_{\text{LoS}}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_h|} \sum_{i \in B_h} [y_i - \text{FSPL}_i - C_{2\text{ray},i}(\hat{\rho}, \hat{\phi})], \quad (13)$$

and the selected pair minimizes the corresponding RMSE. The fitted ρ , ϕ , and β_{LoS} curves are then smoothed across neighbouring height bins, weighted by bin support, and linearly interpolated at inference.

The NLoS branch uses the same training city principle but with a linear morphology calibrator. For every supported regime r , the calibration weights solve

$$\hat{\mathbf{w}}_{\text{NLoS},r} = \arg \min_{\mathbf{w}} \sum_{i \in r} (\mathbf{x}_{\text{NLoS},i}^\top \mathbf{w} - y_i)^2, \quad (14)$$

with fallback to stored antenna height keys within the same morphology and LoS/NLoS group when the exact antenna height key is unavailable. At deployment, the same deterministic features select one stored coefficient vector and produce the clipped NLoS prior in (10). This is an ordinary least squares objective, not a requirement that every regime have a nonsingular normal matrix; if a calibration block is rank deficient or ill conditioned, the stored solution is interpreted as the numerical least squares or pseudoinverse solution associated with this objective.

The delay spread and angular spread priors are fitted in a transformed domain,

$$z = \log(1 + y), \quad \hat{y} = \exp(\hat{z}) - 1, \quad (15)$$

because both standard channel models and the CKM histograms make spread quantities more regular in log space [20], [31]. The released spread maps have a near zero mass plus a long tail: LoS direct path pixels cluster near zero, while blocked or rooftop scattered pixels create rare but important large values. The standardized models justify the log domain

TABLE IV
TERM BY TERM PROVENANCE OF THE MAIN FROZEN PRIOR QUANTITIES.

Term	Prior block	Source	Role in this paper
<i>Literature formula structures</i>			
d_{LoS}	LoS PL	CKM geometry + Friis distance term [16], [17]	Direct transmitter–receiver distance used in FSPL.
d_{ref}	LoS PL	Image source two-ray geometry [16], [17], [21]	Length of the single ground reflected ray.
$C_{2\text{ray}}$	LoS PL	Coherent direct/reflected field sum [16], [17]	Constructive/destructive radial rings before learned correction.
P_0	NLoS PL	COST231/Hata trend [18] + A2G elevation envelope [3], [19]	Raw blocked link scale before morphology OLS calibration.
<i>CKM deterministic geometry and morphology features</i>			
δ_k, h_k, n_k	NLoS PL and spreads	Topology and LoS/NLoS masks	Local density, building height, and NLoS support in a $k \times k$ window.
b_{41}, c_{41}, t_{41}	Spreads	Obstruction features motivated by A2G clutter behavior [3], [32], [33]	Blocker depth, transmitter clearance, and taller building fraction.
$\theta_{\text{norm}}, \theta_{\text{inv}}$	Spreads	Geometry + A2G elevation dependence [3], [19], [20]	Elevation angle and low elevation complement for blocked/spread prone regimes.
<i>CKM training split fitted calibration</i>			
$\rho(h_{\text{tx}}), \phi(h_{\text{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed effective reflection amplitude and phase; not material constants.
$\beta_{\text{LoS}}(h_{\text{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed height dependent offset left after coherent two-ray fitting.
$r(d_{2D}, h_{\text{tx}})$	LoS PL	Train split fit + smoothing	Smoothed deployed radial residual left by two-ray plus bias.
$b_q^{\text{nat}}, b_{\text{top}}, a_j$	Spreads	Fixed CKM raw prior constants	Initial log domain DS/AS scale before ridge calibration.

TABLE V
FROZEN PRIOR ESTIMATORS USED BY THE FINAL MODEL. ALL COEFFICIENTS ARE FITTED ON TRAINING CITIES AND FROZEN BEFORE NEURAL RESIDUAL TRAINING; CITATIONS INDICATE THE LITERATURE BASIS OF EACH STARTING STRUCTURE.

Target/regime	Starting structure	Training city calibration	Deployed prior map
PL LoS	FSPL plus coherent direct/reflected two-ray correction [16], [17], [21]	$\rho(h_{\text{tx}}), \phi(h_{\text{tx}}), \beta_{\text{LoS}}(h_{\text{tx}})$, and radial residual $r(d_{2D}, h_{\text{tx}})$ in smoothed 5 m height bins	LoS channel attenuation map carrying the dominant height/radius rings before any neural residual is added.
PL NLoS	Conservative envelope of COST231 shaped urban loss and an elevation aware A2G-NLoS term [3], [18]–[20]	Per regime OLS over $\mathbf{x}_{\text{NLoS}}$, selected by morphology, visibility, and antenna height group with fallback keys and output clipping	NLoS channel attenuation map that stores blocked link scale, local density, blocker, and elevation effects.
DS / AS	Log domain raw spike plus tail scale from visibility, range, elevation, and local morphology [20], [31]–[33]	Ridge calibration over $\mathbf{x}_{\text{sp}}$ per metric/regime, with height collapsed, topology wide, visibility wide, and global fallbacks	Delay- and angular spread maps in native units after $\exp(\hat{z}) - 1$, used as frozen statistical anchors.

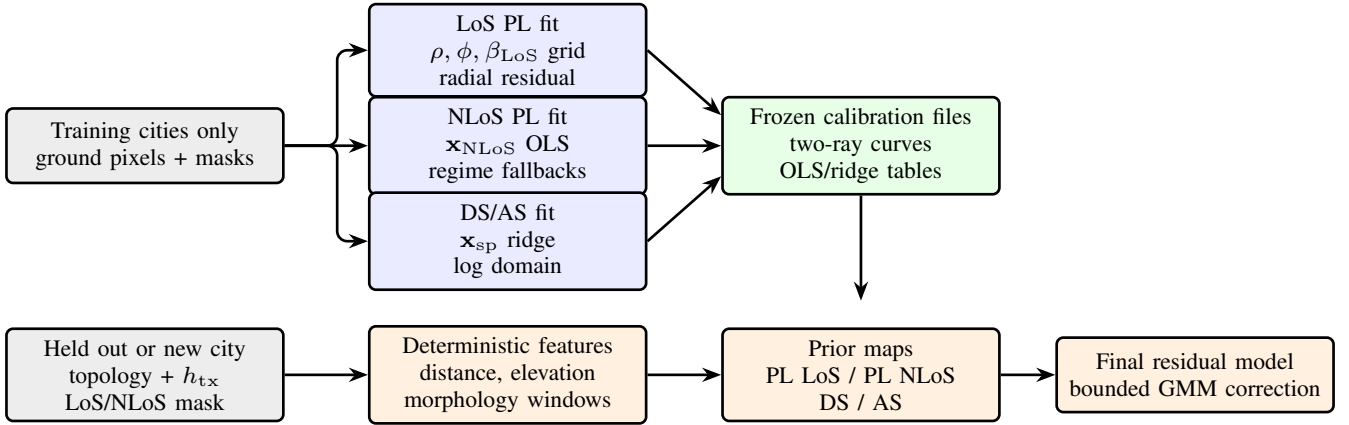


Fig. 3. Prior calibration and deployment. The upper row fits and stores the prior parameters using training cities only. The lower row applies those frozen tables to a held out sample before the neural residual model is called.

large scale variable; the particular range, elevation, density, height, and NLoS support terms below are CKM specific deterministic covariates chosen to make that log variable depend on observable map geometry. Their raw prior is

$$\begin{aligned}
 z^{(0)} = & \log(1 + b_q^{\text{nat}}) + b_{\text{top}} + a_1 \log(1 + d_{2D}) + a_2 \theta_{\text{inv}} \\
 & + a_3 \delta_{41} + a_4 h_{41} + a_5 n_{41} + a_6 n_{41} \theta_{\text{inv}},
 \end{aligned} \quad (16)$$

The constants in (16) are fixed raw prior hyperparameters before ridge calibration. They come from the non neural spread prior implementation rather than from the final residual network. Their role is to give $z^{(0)}$ a physically motivated

starting scale: LoS and NLoS anchors encode the spike plus tail split visible in CKM spread histograms, topology offsets rank broad morphology classes, and positive weights on range, low elevation, density, height, and NLoS support encode the expectation that richer urban multipath increases spread. The ridge fit then learns the CKM specific linear correction around this raw log prior. Numerically, for delay spread, $b_{\text{LoS}}^{\text{nat}} = 4.0 \text{ ns}$, $b_{\text{NLoS}}^{\text{nat}} = 11.0 \text{ ns}$, and $(a_1, \dots, a_6) = (0.11, 0.62, 0.48, 0.32, 0.86, 0.60)$. For angular spread, $b_{\text{LoS}}^{\text{nat}} = 3.0^\circ$, $b_{\text{NLoS}}^{\text{nat}} = 7.0^\circ$, and $(a_1, \dots, a_6) = (0.05, 0.52, 0.60, 0.38, 0.58, 0.35)$. The topology offsets b_{top} are log domain offsets, not native ns/degree offsets. In the

class order open sparse low rise, open sparse vertical, mixed compact low rise, mixed compact mid rise, dense block mid rise, and dense block high rise, the delay spread offsets are (0.00, 0.18, 0.06, 0.22, 0.24, 0.34), and the angular spread offsets are (0.00, 0.09, 0.05, 0.16, 0.18, 0.24). The six topology classes are assigned from map level building coverage $\bar{\delta}_{\text{map}}$ and mean above ground building height $\bar{h}_{\text{map}}$: coverage $\bar{\delta}_{\text{map}} \leq 0.12$ is open sparse, with low rise if $\bar{h}_{\text{map}} \leq 12$ m and vertical otherwise; coverage $\bar{\delta}_{\text{map}} \geq 0.28$ is dense block, with mid rise if $\bar{h}_{\text{map}} \leq 28$ m and high rise otherwise; intermediate coverage is mixed compact, with low rise if $\bar{h}_{\text{map}} \leq 12$ m and mid rise otherwise. These labels are map level routing keys for calibration, not per pixel learned features.

Here q is the LoS/NLoS state. The native anchor b_q^{nat} gives a different starting scale to LoS and NLoS pixels, b_{top} shifts the baseline by topology class, $\log(1 + d_{2D})$ captures range growth, θ_{inv} increases the prior for low elevation links, and δ_{41} , h_{41} , and n_{41} summarize local building density, height, and NLoS support. This raw log prior is intentionally simple; it encodes the expected spike plus tail shape before CKM specific calibration. A 23-term feature vector then augments $z^{(0)}$ with range, elevation, height, multi scale morphology, blocker depth, clearance, and interaction terms:

$$\mathbf{x}_{\text{sp}} = [(z^{(0)})^2, z^{(0)}, \log(1 + d_{2D}), \theta_{\text{norm}}, \theta_{\text{inv}}, h, h^2, \delta_{15}, \delta_{41}, h_{15}, h_{41}, n_{15}, n_{41}, n_{41} \log(1 + d_{2D}), \delta_{41} \theta_{\text{inv}}, b_{41}, 1, c_{41}, t_{41}, \theta_{\text{norm}} \delta_{41}, c_{41} \theta_{\text{inv}}, t_{41} \delta_{41}, t_{41} n_{41}]^T. \quad (17)$$

Here h is normalized transmitter height: one scalar per sample broadcast to each valid receiver row. By contrast, b_{41} , c_{41} , and t_{41} are 41×41 map summaries: local blocker depth, transmitter clearance over nearby buildings, and fraction of locally taller buildings. These height terms are not incidental: they tell the spread prior whether the centred UAV is above the local skyline, near rooftop level, or below nearby high rise clutter. The quadratic raw prior terms let the calibration bend the initial log domain estimate, while the interaction terms represent cases where range, low elevation, dense buildings, and NLoS support become harmful together; in particular, $c_{41} \theta_{\text{inv}}$ means that transmitter clearance is allowed to matter differently at low elevation, not that clearance is only important when NLoS support is already high. The calibrated log prior is ridge regularized per metric and regime:

$$\hat{\mathbf{w}}_{\text{sp},r} = (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{D}^T \mathbf{D})^{-1} \mathbf{X}^T \mathbf{z}. \quad (18)$$

The calibrated log prior is then $\hat{z} = \mathbf{x}_{\text{sp}}^T \hat{\mathbf{w}}_{\text{sp},r}$. The diagonal regularizer $\mathbf{D}$ keeps the calibrated curve close to the raw prior unless the training residuals consistently justify a correction. In practice this matters for the spread targets: a flexible unregularized fit can chase rare high spread pixels, whereas the ridge fit keeps the near zero LoS bulk and the NLoS tail connected to the same physical features.

The spread prior should be read as a statistical prior rather than as a ray by ray multipath model. It does not attempt to reconstruct individual reflections. Instead, it predicts the expected scale of a dense spread map from height, range, LoS/NLoS state, and local morphology. The log transform

TABLE VI
FROZEN PRIOR ONLY RMSE ON THE FINAL HELD OUT TEST SPLIT. THESE ARE THE ANCHORS THAT THE RESIDUAL MODEL MUST IMPROVE.

Target	Overall	LoS	NLoS	Unit
Attenuation	1.9383	1.7519	3.5143	dB
Delay spread	28.1211	27.5119	29.6157	ns
Angular spread	13.8162	15.3751	8.6614	deg

makes the near zero mass and the long tail share one regression space, while the regime fallbacks prevent small calibration subsets from inventing unsupported high spread behaviour.

All calibrations use training cities only. The spread prior dictionary contains 114 fitted regime keys: 36 exact topology/visibility/height keys per metric plus topology, visibility, and global fallbacks. Height is handled differently across priors: the LoS PL branch interpolates 5 m curves, whereas NLoS PL and spread branches use antenna height regimes (`low_ant`, `mid_ant`, `high_ant`) while still keeping continuous height, clearance, and taller building features inside $\mathbf{x}_{\text{sp}}$. The point of this heavy calibration is to store many small decisions about morphology, visibility, and height before the neural residual is trained. On the final held out test split, the deployed prior anchors are summarized in Table VI. Beyond the topology, masks, and scalar height path, the final neural model receives the deployed prior maps rather than the handcrafted feature vectors; those features remain inside the frozen prior estimators instead of becoming additional learned inputs.

C. HARP-Net CKM Residual Model

The neural model receives nine aligned channels: topology, LoS mask, NLoS mask, ground mask, combined path loss prior, LoS path loss prior, NLoS path loss prior, delay spread prior, and angular spread prior. Height is encoded with sinusoidal features and injected through FiLM modulation [37], so the final network can adapt the residual correction continuously between the calibrated height regimes instead of learning one height agnostic correction. The backbone is a shared U-Net style encoder decoder with GroupNorm, FiLM conditioning, and task specific residual heads.

The actual final model is therefore not a replacement for the priors. Its input tensor already contains the deployed attenuation and spread estimates, and its output head is constrained to make bounded corrections around them. This choice changes the learning problem: the encoder decoder does not need to infer absolute free space attenuation, altitude trends, or the average spread scale from pixels alone. It learns where the frozen estimates are locally biased and how the remaining residual distribution should be placed across the map.

The height path mirrors the full implementation. The scalar h_{tx} is expanded to a 32-dimensional sinusoidal code using 16 log spaced frequencies over the transmitter height range,

$$\mathbf{e}(h_{\text{tx}}) = \left[\sin \frac{h_{\text{tx}}}{f_i}, \cos \frac{h_{\text{tx}}}{f_i} \right]_{i=1}^{16} \in \mathbb{R}^{32}, \quad (19)$$

then mapped by an MLP to a $\mathbb{R}^{160}$ condition vector. Each FiLM site projects this vector to channel wise scale and shift

parameters and modulates a feature map as

$$\mathbf{X}' = \mathbf{X} \odot (1 + \mathbf{g}) + \mathbf{b}. \quad (20)$$

Encoder FiLM uses the height condition directly; decoder FiLM and the global task heads use a fused condition formed from the height code and the pooled bottleneck context. Thus the same centred topology can receive different residual corrections when the UAV is at 30 m, 150 m, or 350 m. The height aware results in Sec. V show that this is not only an architectural convenience: sinusoidal height embeddings with FiLM modulation are sufficient for one model to cover the full varying height range without training a separate network per altitude bin.

The head uses a two branch residual logic around the frozen prior. The global branch asks: “what residual distribution does this whole map, height, and topology imply?” The local branch asks: “where should each residual component be placed in the 513×513 map?” This is the key distinction from a direct U-Net regressor. The network does not directly overwrite the prior with a free form map; it predicts a small residual distribution and then assigns that residual pixel by pixel.

For each task k , visibility region $r \in \{\text{LoS}, \text{NLoS}\}$, component $j \in \{1, 2, 3\}$, and pixel p , the global head receives the pooled bottleneck and fused height/context vector and outputs

$$\{\pi_{k,r,j}^{(g)}, \delta_{k,r,j}^{(g)}, \sigma_{k,r,j}^{(g)}\}_{j=1}^3.$$

After the implementation transforms, $\pi^{(g)}$ is a map level residual mixture distribution, $\delta^{(g)}$ is a bounded global residual offset, and $\sigma^{(g)}$ is a bounded global uncertainty scale. The local head receives the full resolution decoder feature map and outputs

$$\{p_{k,r,j}(p)\}_{j=1}^3, \quad \delta_{k,r}^{(\ell)}(p), \quad \alpha_{k,r}(p), \quad \tilde{\sigma}_{k,r}(p).$$

Here $p_{k,r,j}(p)$ is the per pixel component assignment, the local residual $\delta^{(\ell)}$ shifts all components at that pixel, α is an anchor gate initialized with bias -2 so the model starts close to the prior, and $\tilde{\sigma}$ is the local uncertainty term. This extends the distribution aware line of work from scalar mixture predictors to dense residual maps [27]–[29]. The native residual is clipped by target and visibility:

$$\Delta_{k,r,j}(p) = \text{clip}\left(\delta_{k,r,j}^{(g)} + \delta_{k,r}^{(\ell)}(p), -\delta_{\max,k,r}, \delta_{\max,k,r}\right), \quad (21)$$

with caps of 7/25 dB for attenuation (PL) LoS/NLoS, 20/20 ns for DS, and $17^\circ/15^\circ$ for AS. The residual cap is specified in native units: dB for attenuation (PL), ns for DS, and degrees for AS. Omitting indices for compactness, the component mean is anchored to the frozen prior after this native residual has been assembled:

$$\begin{aligned} u^{\text{nat}} &= y^{\text{prior,nat}} + \alpha \Delta, \\ \mu &= u^{\text{nat}}, \quad k = \text{PL}, \\ \mu &= \log(1 + [u^{\text{nat}}]_+), \quad k \in \{\text{DS}, \text{AS}\}. \end{aligned} \quad (22)$$

where $[u^{\text{nat}}]_+ = \max(u^{\text{nat}}, 0)$ and $0 \leq \alpha_{k,r}(p) \leq 1$. The attenuation branch (PL) is modeled in native dB space, while DS and AS use the log transform in (15). The deterministic

map used for RMSE is the mixture mean in the modelled space:

$$\hat{z}_{k,r}(p) = \sum_{j=1}^3 p_{k,r,j}(p) \mu_{k,r,j}(p). \quad (23)$$

For the attenuation branch (PL), $\hat{y}_{k,r} = \hat{z}_{k,r}$. For DS and AS, the output is mapped back with $\hat{y}_{k,r} = \exp(\hat{z}_{k,r}) - 1$. Thus the spread output is the deterministic point map used for RMSE, not the exact native space expectation of the log space mixture. The final map selects the LoS or NLoS branch with the explicit geometric mask. The mixture variances combine map level and local uncertainty,

$$\sigma_{k,r,j}(p) = \sqrt{\left(\sigma_{k,r,j}^{(g)}\right)^2 + \left(\tilde{\sigma}_{k,r}(p)\right)^2}, \quad (24)$$

with both terms constrained to $[0.05, 3.0]$. During the probabilistic residual phase these variances and mixture weights define a pixel wise likelihood; during the final fine tuning phase the same head remains in place but the reported output is the deterministic point map above.

The two mixture weight objects have different jobs. The global weights $\pi_{k,r,j}^{(g)}$ describe the sample level residual histogram and are used by the distributional regularizer. The local weights $p_{k,r,j}(p)$ are the pixel wise assignment probabilities used in the final dense map. During the probabilistic residual phase the per pixel density is

$$q_{k,r}(z_k(p) | p) = \sum_{j=1}^3 p_{k,r,j}(p) \mathcal{N}(z_k(p); \mu_{k,r,j}(p), \sigma_{k,r,j}^2(p)), \quad (25)$$

and the regional NLL is the masked average of $-\log q_{k,r}(z_k(p) | p)$ over valid LoS or NLoS pixels. Separately, the KL term bins the true residuals $\epsilon_k(p) = y_k(p) - y_k^{\text{prior}}(p)$ into a soft empirical histogram and compares it with the histogram implied by $\{\pi_{k,r,j}^{(g)}, \delta_{k,r,j}^{(g)}\}$. This is what makes the head “distribution first”: before the decoder places values pixel by pixel, the global branch is pressured to put mixture mass where the map residuals actually live.

This head is useful even when the final reported number is a deterministic RMSE. The probabilistic phase teaches the network that a hard NLoS region can contain several plausible residual levels, while the anchor gate decides how far the prediction may move from the frozen estimate. The final point map is therefore not an unconstrained average of arbitrary map values; it is the model space mean of a small residual mixture whose components remain tied to the prior.

HARP-NET CKM should not be read as only the final fine tuning loss. It has two active training phases: a large capacity probabilistic residual phase with probabilistic losses enabled, followed by an RMSE/MAE dominant fine tuning phase. The composite objective implemented by the model is

$$\begin{aligned} \mathcal{L} &= \sum_k w_k (\lambda_{\text{NLL}} \mathcal{L}_{\text{NLL},k} + \lambda_{\text{KL}} \mathcal{L}_{\text{KL},k} + \lambda_{\text{mom}} \mathcal{L}_{\text{mom},k} \\ &\quad + \lambda_{\text{anchor}} \mathcal{L}_{\text{anchor},k} + \lambda_{\text{guard}} \mathcal{L}_{\text{guard},k} + \lambda_{\text{out}} \mathcal{L}_{\text{out},k} \\ &\quad + \lambda_{\text{RMSE}} \mathcal{L}_{\text{RMSE},k} + \lambda_{\text{MAE}} \mathcal{L}_{\text{MAE},k}). \end{aligned} \quad (26)$$

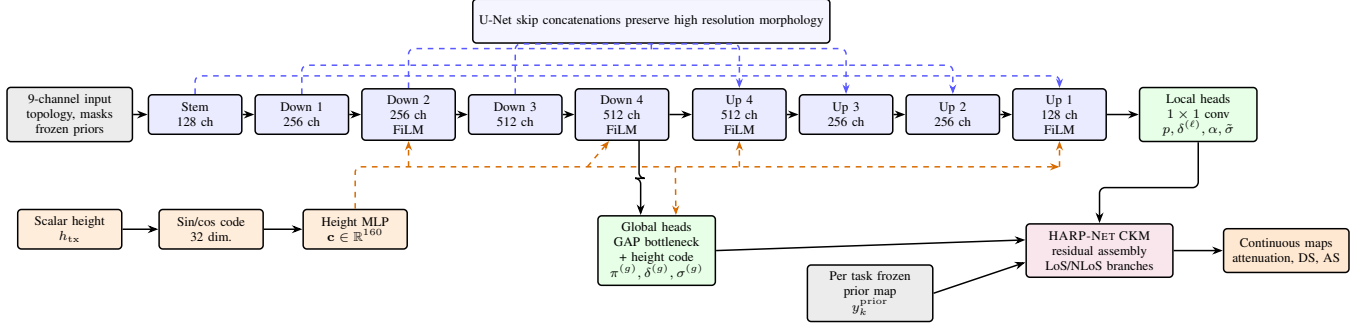


Fig. 4. Final shared deep learning architecture used in the reported system diagram. The map stack carries topology, masks, and frozen priors; the scalar Transmitter height is encoded separately and injected through FiLM. Encoder FiLM uses the height code; decoder FiLM and global heads use the fused height/bottleneck context. A shared U-Net produces local residual fields, pooled bottleneck features produce global GMM parameters, and the HARP-NET CKM residual assembly emits continuous channel attenuation, DS, and AS maps. The blue dashed arrows show the four encoder-decoder concatenations used in the implementation.

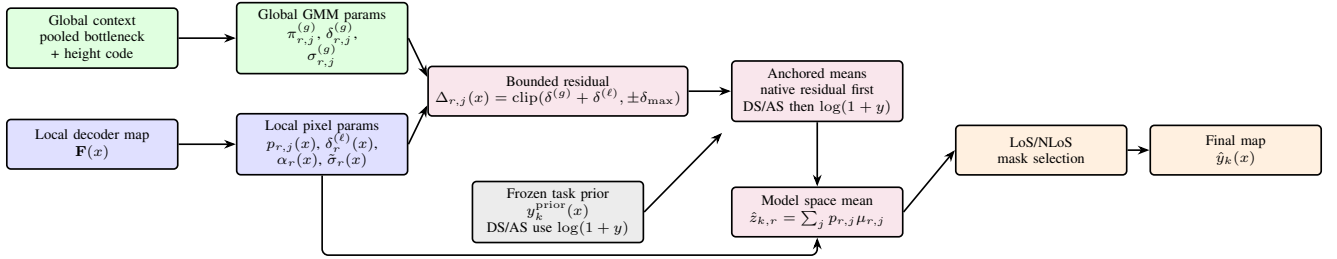


Fig. 5. HARP-NET CKM residual head. The global branch predicts the sample level residual mixture; the local branch places that distribution pixel by pixel through component weights, residual refinements, gates and local uncertainty. Component means are anchored to the frozen prior and the deterministic map is obtained from the pixel wise model space mixture mean after LoS/NLoS selection.

The probabilistic terms matter during the probabilistic residual phase. The NLL term teaches the head to place mixture mass around the transformed target, the KL and moment terms keep the mixture distribution from becoming a numerically convenient but physically uninformative density, and the outlier budget term discourages a solution that spends most of its capacity on the easy bulk while leaving large tails unconstrained. The final fine tuning phase then disables those terms and optimizes the deterministic mixture mean point map for RMSE/MAE, with a small anchor and guard that keep the correction close to the prior. Table VII summarizes the roles. The selected checkpoint is therefore reported as a deterministic RMSE improvement over the frozen priors, not as a calibrated probabilistic predictor. In this final phase, checkpoint selection uses the validation composite $PL_{RMSE} + 0.30 DS_{RMSE} + 0.25 AS_{RMSE}$, matching the reported `score_weights`.

The practical role of the two training phases is simple. The first phase makes the head behave like a residual density model: it learns where residual mass lies and discourages the old single mode collapse. The final fine tuning phase keeps the same architecture but shifts the optimization weight toward the deterministic metrics used for the paper tables. The anchor and guard terms remain active at small weight so the fine tuned model cannot gain average RMSE by damaging large regions where the frozen prior was already accurate.

IV. EXPERIMENTS AND RESULTS

A. Main Test Results and City Audit

Before collapsing the test split to one average, Table X shows the city level audit. Training cities are omitted because

the priors and the residual model were fitted on them; validation cities are used for model selection and are not part of the final claim. The test rows show that the aggregate result is not a single easy city effect. Osaka, Kuala Lumpur, Phuket, Barcelona, and Pune are harder for delay spread, while Carcassonne, Halifax, Key West, and Vancouver are easier under this split. The topology and height mix columns explain part of this variation: cities with more low or mid height samples and denser morphology generally show larger spread errors.

Table XI then collapses the same held out test split to the target averages. The model reduces RMSE relative to the frozen prior for all three targets. Channel attenuation is the cleanest success case because both RMSE and MAE improve. Angular spread also improves strongly in RMSE and remains approximately neutral in MAE. Delay spread improves RMSE while MAE slightly increases; this is consistent with reducing rare high delay errors while introducing small local shifts.

The original numerical targets were 5 dB, 50 ns, and 20°. The final model clears all three on unseen cities. More importantly, it does so jointly: the same network predicts channel attenuation, DS, and AS instead of using separate specialist models. The targets are broad acceptance thresholds rather than equally informative final benchmarks. In particular, the attenuation prior is already below 2 dB, so the final attenuation question is whether the residual network beats that prior and preserves map structure; Tables XI and XII show that it does. The map correlation results in Table XII add a structural check: after each map's mean and scale are removed, the residual model places the relative high and low regions better

TABLE VII
LOSS TERMS AVAILABLE IN THE HARP-NET CKM OBJECTIVE AND THEIR PRACTICAL PURPOSE.

Term	Active phase	Role
GMM NLL	probabilistic residual phase	Fits the per pixel mixture density in native attenuation (PL) space and log transformed spread space.
Distribution KL	probabilistic residual phase	Regularizes the learned mixture so it remains close to the intended distributional structure instead of using degenerate weights or variances.
Moment matching	probabilistic residual phase	Encourages the model space mean and spread of the mixture to agree with target moments, stabilizing deterministic extraction from the GMM.
Outlier budget	probabilistic residual phase	Keeps rare large residuals visible in the objective instead of letting the global average hide them.
Anchor	both phases	Penalizes unnecessary deviation from the frozen prior; this is especially important because the prior is already a strong calibrated predictor.
Prior guard	both phases	Adds a penalty when the model becomes worse than the prior at valid pixels, discouraging destructive corrections.
RMSE / MAE	both phases, dominant in final fine tuning	Optimizes the deterministic map reported in the final evaluation; RMSE handles tails and MAE tracks typical absolute error.

TABLE VIII
TRAINING CONFIGURATION OF THE FINAL LINE.

Item	Value
Image size	513×513
Input channels	9 map channels plus height FiLM conditioning
Backbone	GroupNorm U-Net, base width 128, FiLM dim. 160
Residual head	3 mixture components, bounded residuals
Optimizer	AdamW, learning rate 5×10^{-5} , weight decay 0.01
Batching	batch size 1, gradient accumulation 8
Scheduler	1 warmup epoch plus cosine LambdaLR ; early stopping patience 10
Augmentation	Training D4 rotations/reflections only; evaluation uses no augmentation
Checkpoint selection	Validation composite PL RMSE +0.30 DS RMSE +0.25 AS RMSE
Loss schedule	Large capacity probabilistic residual phase plus final residual fine tuning; see Table IX

TABLE IX
HARP-NET CKM LOSS SCHEDULE. THE PRE FINE TUNING PROBABILISTIC RESIDUAL PHASE IS PART OF THE FINAL PIPELINE, EVEN THOUGH THE REPORTED CHECKPOINT IS SELECTED AFTER THE DETERMINISTIC FINE TUNING PHASE.

Setting	Probabilistic residual phase	Final fine tuning
NLL / KL / moment / outlier	1.00 / 0.25 / 0.10 / 0.02	0 / 0 / 0 / 0
RMSE / MAE	1.50 / 0.10	1.00 / 0.05
Anchor / guard	0.05 / 0.10	0.01 / 0.02
Task weights	PL 1.0, DS 1.0, AS 1.0	PL 1.0, DS 0.25, AS 0.25
Learning rate	2×10^{-4}	5×10^{-5}

than the frozen priors for all three targets. This is especially important for the spread maps. Delay spread RMSE improves by about 5% and angular spread RMSE by about 17%, but their MapCorr values rise by about 36% and 67% relative to the frozen priors. Thus the residual model is doing visible map-structure correction even when the unit aware RMSE gain looks modest.

a) Channel attenuation: The frozen path loss prior is already strong, reaching 1.9383 dB RMSE and 1.2319 dB MAE on the final test split. HARP-NET CKM reduces these to 1.6947 dB and 0.9772 dB. This is the most uniform gain: the residual model improves both squared error and typical absolute error, and it improves both LoS and NLoS pixels in Table XIII. The closest external references are dense path loss radio map papers, but they still differ in split, height protocol, frequency, and target definition.

b) Angular spread: The angular spread prior reaches 13.8162° RMSE and 5.0033° MAE; the final model reaches 11.5253° RMSE and 4.8899° MAE. The MAE gain is smaller

than the RMSE gain, so the residual model mainly suppresses larger angular spread errors rather than shifting every pixel uniformly. The LoS gain is much larger than the NLoS gain, which is expected because the NLoS angular spread prior is already close to the target distribution. No reviewed external method reports the same dense CKM city holdout angular spread map task. The nearest papers either predict link level or sparse angular spread channel parameters [23], [24], or reconstruct channel characteristic maps from lower resolution channel maps [25]. The primary numerical comparison is therefore against the frozen prior and the same protocol target.

c) Delay spread: The delay spread prior reaches 28.1211 ns RMSE and 7.3041 ns MAE; the final model reaches 26.6882 ns RMSE and 7.8160 ns MAE. This is the least uniform target: RMSE improves because rare high delay errors are reduced, while MAE slightly worsens because small local shifts appear in many easy pixels. The LoS/NLoS split confirms that the model improves both visibility cases in RMSE. Similar A2G delay spread predictors exist [22], [23], and Wang et al. provide a map based super resolution reference [25]. None is the same topology to dense CKM city holdout generation task, so they are used as context rather than as direct leaderboards.

B. LoS/NLoS and Split Diagnostics

Table XIII separates valid pixels by the geometric visibility mask. The residual model improves the prior in both LoS and NLoS pixels for all targets. This is important because global RMSE could otherwise hide a failure in the physically hard NLoS subset.

The validation/test sanity check is also useful. The validation row evaluates the selected checkpoint on validation cities, while the test row is the real generalization result. Delay spread worsens more on test because the held out city mix contains harder high delay cases. The frozen delay spread prior shows the same validation to test worsening, confirming that this is a city mix effect rather than an evaluator change.

The antenna breakdown in Table XV is a held out test audit, not a training result. It shows that all three height bins improve for all targets. Low altitude remains the hardest spread regime, while high altitude is easier for DS and AS because fewer rays are trapped by local rooftops and street canyons.

TABLE X
HELD OUT TEST RMSE BY CITY. ATTENUATION IS IN DB, DELAY SPREAD IN NS AND ANGULAR SPREAD IN DEGREES. TOPOLOGY MIX IS OPEN/MIXED/DENSE; HEIGHT MIX IS LOW/MID/HIGH ANTENNA PERCENTAGE.

City	Samples	Atten.	DS	AS	Topology mix	Height mix
Barcelona	50	1.74	33.0	11.1	30/30/40	40/32/28
Beijing	200	1.57	23.3	10.7	28/65/8	30/41/30
Carcassonne	110	1.43	8.0	9.7	41/50/9	30/29/41
Copenhagen	220	1.83	27.3	12.0	14/68/18	37/27/36
Halifax	250	1.43	19.9	9.7	78/22/0	36/33/31
Jaipur	120	2.05	21.6	11.9	12/25/62	37/34/29
Johannesburg	245	1.76	21.0	12.1	10/55/35	34/34/32
Key West	105	1.39	20.0	10.4	52/43/5	33/30/37
Kuala Lumpur	360	2.05	44.2	13.5	18/44/38	29/38/34
Osaka	50	2.43	37.3	14.9	0/60/40	28/44/28
Phuket	50	1.82	37.2	13.8	20/50/30	28/36/36
Pune	135	1.95	33.3	13.2	15/67/19	32/27/41
Surat	190	1.63	23.5	10.3	34/16/50	36/32/32
Vancouver	505	1.53	17.7	10.8	29/67/4	31/35/34

TABLE XI
FINAL MODEL TEST RESULTS ON UNSEEN CITIES. NEGATIVE Δ MEANS THAT THE RESIDUAL MODEL IMPROVES THE FROZEN PRIOR.

Target	Model RMSE	Prior RMSE	Δ RMSE	Model MAE	Unit
Attenuation	1.6947	1.9383	-0.2436	0.9772	dB
Delay spread	26.6882	28.1211	-1.4329	7.8160	ns
Angular spread	11.5253	13.8162	-2.2909	4.8899	deg

TABLE XII
FINAL MODEL TEST MAP CORRELATION ON UNSEEN CITIES. HIGHER IS BETTER.

Target	Model	Prior	Δ
Attenuation	0.961055	0.947710	+0.013345
Delay spread	0.414692	0.306026	+0.108666
Angular spread	0.581783	0.348713	+0.233070

TABLE XIII
FINAL TEST RMSE SEPARATED BY LOS AND NLOS PIXELS.

Target	Scope	Model	Prior	Δ
Atten.	LoS	1.5175	1.7519	-0.2344
Atten.	NLoS	3.1596	3.5143	-0.3547
DS	LoS	25.6137	27.5119	-1.8983
DS	NLoS	29.2464	29.6157	-0.3693
AS	LoS	12.5270	15.3751	-2.8481
AS	NLoS	8.4550	8.6614	-0.2064

C. Prior and Split Breakdowns

The frozen priors are strong enough to deserve their own reporting. They are not simply baselines for the final neural model; they encode most of the stable geometry and therefore determine what residual problem the network sees. Because these prior calibrations are deterministic, non DL estimators, their standalone audits are reported separately from neural checkpoint selection. The path loss table uses the held out validation+test split after fitting the 10840 training references and evaluating 5340 held out references (4994 topology assigned rows). The spread table uses the final 14-city test split because the calibrated spread prior code and the final residual model use the same 10840/2750/2590 train/validation/test split. Table XVI shows the height dependence of the calibrated path loss prior. This is not post hoc stratification: transmitter height is part of the prior estimator through the centred geometry, elevation angle, and height binned calibration, and it later conditions the residual neural model through FiLM. The trend is strongest in NLoS: the low altitude bin remains the hardest

TABLE XIV
VALIDATION VERSUS HELD OUT TEST CHECK FOR THE SAME SELECTED MODEL.

Split	Samples	Atten. RMSE	DS RMSE	AS RMSE
Validation	2750	1.6272	22.1946	11.4141
Test	2590	1.6947	26.6882	11.5253

because many receivers are blocked by local urban structure, while high UAVs see fewer deep shadow pockets. Table XVII shows the matching spread prior pattern. Delay spread in particular collapses with altitude, from 39.90 ns overall in the lowest bin to 3.70 ns above 300 m. Combined with the empirical HDF5 histogram in Fig. 1, this means the high altitude rows are not better because they have more training support; the training split contains only 866 samples above 300 m. They are better despite that scarcity, which indicates a physically easier regime with fewer blocked links and weaker spread tails.

The path loss prior also varies by topology. Its overall RMSE ranges from 1.43 dB in open sparse low rise maps to 2.97 dB in dense block high rise maps. The NLoS branch is hardest in sparse vertical and dense high rise regimes, reaching 3.91 dB and 4.09 dB; the overall pixel weighted prior remains 1.92 dB.

The residual model is therefore reported on held out cities rather than on the training split. The validation row in Table XIV is kept only as a checkpoint selection sanity check, while the test row is the generalization claim. Training RMSE is deliberately omitted from the paper because it does not measure transfer to unseen city morphologies.

The same principle is used for detailed residual audits: topology and transmitter height breakdowns should be read on validation or test cities, not on the full training plus held out pool. The paper keeps a compact test city audit in Table X, the prior only audits above, and the held out test summaries in Tables XI–XIV. Finer cross products of topology, height and

TABLE XV
HELD OUT TEST RMSE BY ANTENNA HEIGHT BIN. EACH Δ COLUMN IS MODEL MINUS FROZEN PRIOR, SO NEGATIVE IS BETTER.

Antenna bin	Samples	PL	Δ PL	DS	Δ DS	AS	Δ AS
low	846	1.8755	-0.2645	35.33	-2.02	13.15	-2.36
mid	878	1.7382	-0.2527	27.70	-1.40	11.92	-2.54
high	866	1.4990	-0.2165	11.78	-0.46	8.81	-2.08

TABLE XVI
CALIBRATED PATH LOSS PRIOR RMSE BY TRANSMITTER HEIGHT ON THE
HELD OUT VALIDATION+TEST SPLIT.

Height bin	Overall	LoS	NLoS	Samples
12–50 m	2.180	1.797	3.756	1211
50–150 m	1.936	1.785	3.228	2634
150–300 m	1.680	1.666	2.259	786
300–500 m	1.627	1.619	2.255	362

TABLE XVII
CALIBRATED SPREAD PRIOR RMSE BY TRANSMITTER HEIGHT ON THE
FINAL 14-CITY TEST SPLIT.

Height bin	Metric	Overall	LoS	NLoS
12–50 m	DS	39.90	43.04	35.13
50–150 m	DS	27.17	26.72	28.29
150–300 m	DS	8.70	8.79	8.13
300–500 m	DS	3.70	3.80	2.76
12–50 m	AS	15.69	18.81	9.93
50–150 m	AS	14.19	15.89	8.41
150–300 m	AS	10.39	11.07	4.49
300–500 m	AS	8.85	9.34	2.85

visibility are left to the thesis and evaluation artifacts.

D. Metric Interpretation

The reported model is deterministic at evaluation time: all tables compare the continuous HARP-NET CKM point maps with the stored ray-traced target maps over valid ground receiver pixels. The residual head is distribution aware during training because attenuation, delay spread and angular spread have different LoS/NLoS regimes and tail behaviour, but the final claim is the RMSE and MAE of the exported point prediction.

The three RMSE units have different physical meanings. For attenuation, an error of e dB corresponds to a received power factor

$$G(e) = 10^{e/10}. \quad (27)$$

Thus the final 1.6947 dB attenuation RMSE is about $1.48\times$ in received power, while the original 5 dB target would allow a $3.16\times$ factor. A 5 dB error is therefore not small near a coverage edge: it can change outage, link margin, MCS or handover decisions. The final attenuation error is much smaller and also improves a strong 1.9383 dB frozen prior.

For delay spread, time can be converted into an equivalent excess propagation path length,

$$\Delta d = c \Delta \tau, \quad c = 0.299792458 \text{ m/ns}. \quad (28)$$

The final 26.6882 ns delay spread RMSE therefore corresponds to 8.00 m of equivalent multipath path length uncertainty; the original 50 ns target corresponds to 14.99 m. This does not mean that every ray path is geometrically wrong by 8 m. It only gives the physical scale of the remaining RMS temporal

dispersion error. 3GPP TR 38.901 uses example CDL/TDL delay scaling values from very short 10 ns and short 30 ns to 100 ns, 300 ns and 1000 ns [20]; the final DS RMSE is therefore around the short profile scale, but the moderate MapCorr shows that spatial delay placement is still the hardest spread behaviour.

For angular spread, the LoS/NLoS split is more informative than the global average. LoS angular spread can be dominated by the direct ray plus narrow secondary components, so angular spikes may affect RMSE without being the most important link level ambiguity. NLoS angular spread is more relevant for beam selection, spatial richness and multipath ambiguity; the final NLoS AS RMSE is 8.4550° , below both the global 11.5253° value and the original 20° target.

For spread targets, quantized labels make visual and RMSE interpretation harder. The released delay and angular maps are integer valued; smooth predictions can be penalized across staircase like target boundaries even when the morphology is plausible. This does not invalidate RMSE, but it warns against overinterpreting small local residuals. Map correlation is therefore reported as a companion metric: it is insensitive to mapwise bias and scale, so it is useful for checking spatial pattern recovery in DS/AS maps where MAE is less conclusive.

E. Qualitative Residual Behavior

Fig. 6 is the qualitative check. The model receives the frozen prior and predicts a correction. The useful visual evidence is not merely that the prediction resembles the target; it is that the correction row tends to correlate negatively with the prior error row, which is the expected behavior of a HARP-NET CKM residual corrector.

F. External Metric Context

Table XVIII places the final channel attenuation result beside the closest available path loss side references. The table is deliberately compact and cautious: datasets, height assumptions, target definitions, and splits differ. No scalar indoor or link level spread predictor is used as a direct dense map leaderboard.

Table XIX gives the corresponding spread side context. These rows are not direct rankings, because the external works report scalar or link level channel parameters rather than dense CKM maps. They are included so that the DS/AS results are still positioned against the nearest published spread prediction literature.

The fair conclusion is narrow: the attenuation result is in the same range as strong dense radio map references, while spread results should be read mainly against same protocol internal baselines because no external dense topology conditioned

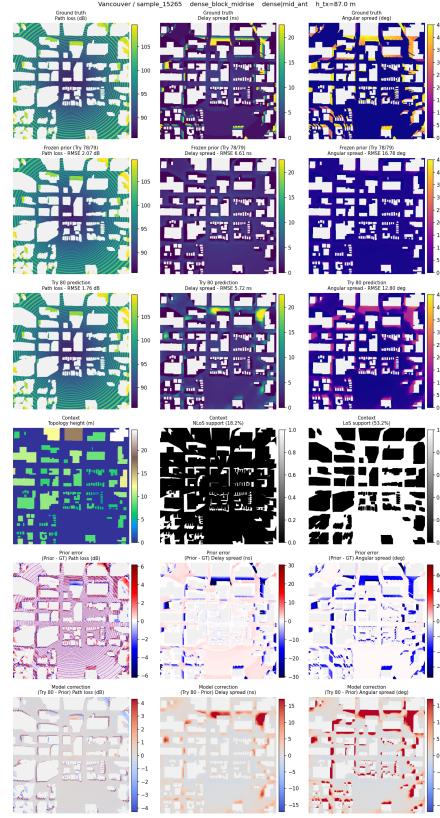


Fig. 6. Representative final model inspection panel for a dense, mid antenna held out test sample in Vancouver. Columns are channel attenuation, delay spread, and angular spread. Rows show ground truth, frozen prior, final prediction, context/support, prior error, and model correction.

TABLE XVIII
COMPACT COMPARISON WITH NEARBY PATH LOSS SIDE METRICS.

Reference	Attenuation / PL side metric	Reading / caveat
RadioUNet [5]	1.6 dB–3.1 dB equivalent PL RMSE	Final attenuation result is in this band; no transmitter height or spread targets.
RadioGUNet [10]	1.304 dB DPM; 1.936 dB IRT	Final attenuation result is between those values; outdoor RadioMapSeer setting.
PMNet / ICASSP [7], [8]	approx. 1.38 dB converted score	Final attenuation result is close but higher under a different benchmark setting; scale context only, no joint spread prediction.
Gao et al. [12]	5.59 dB ICASSP 2023; 12.19 dB ITU 2024	Final attenuation result is numerically lower, but this is scale context rather than a like for like win: recent corridor weighted outdoor PL predictor, no variable height UAV or joint spread outputs.
ReVeal [15]	1.95 dB outdoor sparse measurement RMSE	Final attenuation result is in the same range; sparse mapping rather than dense generation.
RMTransformer / PathFinder [9], [11]	≈ 1.82 dB / ≈ 1.19 dB converted references	Useful scale checks, but random/specialized splits and different assumptions.
AIRMap [13]	Path gain error < 4 dB against ray tracing	Same broad digital twin motivation, but path gain and a fixed 200×200 input grid with variable physical extent.
Final model (ours)	1.6947 dB attenuation; 26.6882 ns DS; 11.5253° AS	Strict CKM city holdout, centred variable height UAV, dense 513×513 channel attenuation/DS/AS.

CKM attenuation/DS/AS city holdout benchmark was found. Wang et al. are the closest map based spread reference, but their task is channel map super resolution rather than topology

to dense CKM generation.

G. Runtime

Runtime was measured on Barcelona layouts at full 513×513 resolution on the same laptop: an AMD Ryzen 5 5600H CPU and an NVIDIA RTX 3050 Ti Laptop GPU. After checkpoint loading, the generator includes mask generation, frozen priors, final prediction, and numeric export. It runs in 0.88 s per sample versus 99.89 s for the local MATLAB ray tracing path, a $113.7\times$ speedup under this setup.

H. Interpretation and Limitations

The final result supports a hybrid CKM design rather than a claim that one backbone solves radio mapping in general. The residual model improves a strong frozen prior on held out validation and test cities, making the gain harder to dismiss as one easy city mix. The modest delay spread gain is also informative: after the calibrated prior absorbs the large geometry trend, remaining spread error is dominated by rare tails, target quantization, and local blocked/scattered paths.

Limitations remain. The labels come from ray tracing rather than field measurements, so the result is simulator agreement, not over the air validation. Target quantization also limits sub unit learning even though the model outputs continuous channel attenuation, DS, and AS values; the 0.9772 dB final attenuation MAE is already close to the 1 dB label step. Finally, the model is selected under one city holdout split and one final seed, with fixed frequency, antenna pattern, and simulator assumptions.

TABLE XIX

SPREAD SIDE COMPARISON WITH NEARBY CHANNEL PARAMETER PREDICTION WORK. EXTERNAL ROWS GIVE TASK AND METRIC CONTEXT, NOT A DIRECT DENSE MAP LEADERBOARD.

Reference	Spread side reported output	Relation to this work	Why it is not direct
Yang et al. [22]	A2G mmWave delay spread prediction with RF/kNN; lower RMSE than a lognormal contrast model.	Closest early A2G delay spread ML precedent.	Scalar DS only; no AS, no dense map, no CKM city holdout.
Huang et al. [23]	Transformer UAV predictor; merged data RMSE 4.577 ns for RMS DS and 1.549° , 0.375° , 0.864° , 0.074° for RMS AAOA, EAOA, AAOA and EAOD.	Recent UAV mmWave model with delay and angular spread outputs.	Per link characteristics from coordinates/frequency/location tags, not dense 513×513 AS/DS maps from topology images.
Mi et al. [24]	Regression PointNet for indoor 60GHz measurements; predicts RMS DS, azimuth AS, PL and K -factor; average AS RMSE 7.39° .	Measurement based DL reference including DS and AS.	Indoor point cloud task with link level samples, not outdoor UAV CKM generation.
Wang et al. [25]	Multitask super resolution of path loss, DS, RMS azimuth/elevation AS and LoS/NLoS maps from lower resolution channel maps.	Closest map based spread reference reviewed.	Super resolution starts from a sampled coarse channel map; this paper predicts dense maps from topology, priors and UAV height with no coarse DS/AS input.
Final model	26.6882 ns DS RMSE and 11.5253° AS RMSE.	Dense channel attenuation/DS/AS maps under the final city holdout setting.	Should be judged mainly against the frozen spread priors and thesis targets under the same protocol.

TABLE XX

RUNTIME COMPARISON FOR ONE FULL RESOLUTION BARCELONA CKM SAMPLE ON THE SAME RYZEN 5 5600H / RTX 3050 Ti LAPTOP HARDWARE.

Method	Timed path	Time
MATLAB RT	Pixel receivers outside buildings; attenuation/DS/AS parsed from paths; CPU path	99.89 s
Final generator	Masks, priors, attenuation/DS/AS prediction, numeric export; CUDA mixed precision	0.88 s

V. CODE AVAILABILITY

The reproducibility record is public as source repositories. The standalone CKM Generator repository contains the interface, command line generator, frozen calibration files, runtime code, and published checkpoint path [38]. The final training and evaluation repository is the source record for the paper metrics [39], tied to the selected `best_model.pt` checkpoint and exported breakdown JSON files. The LaTeX thesis source and project archive are also public [40], [41]; large data and model artifacts are external artifacts or Git LFS objects.

VI. CONCLUSION

This paper presents a dense variable height UAV CKM surrogate that jointly predicts channel attenuation, delay spread, and angular spread on unseen cities. Under strict city holdout, the final model reaches 1.6947 dB, 26.6882 ns, and 11.5253° , respectively. Its main contribution is a hybrid recipe: fit strong priors on training cities for predictable structure, then learn bounded residual distributions around those anchors.

The calibrated priors combine coherent two-ray LoS path loss, height dependent calibration, morphology aware NLoS correction, visibility switching, and log domain DS/AS spread statistics. These anchors explain most radial, visibility, height, and spread scale structure; the residual model improves them by 0.2436 dB, 1.4329 ns, and 2.2909° RMSE on the held out test set. It also improves map correlation from 0.947710 to 0.961055 for attenuation, from 0.306026 to 0.414692 for delay spread, and from 0.348713 to 0.581783 for angular spread. For the two spread targets this is a relative MapCorr gain of about 36% and 67%, respectively, both above the 25% mark. This makes clear that the residual model improves map similarity

even when the RMSE reduction is moderate. Continuous transmitter height is handled through both the priors and a sinusoidal FiLM path, avoiding per altitude retraining.

The distribution first diagnosis remains the key modelling lesson. NLoS attenuation residuals are mixture like and prone to mode collapse, while DS/AS targets are quantized and spike plus tail. GMM style residual heads and histogram checks prevent visually plausible maps from hiding collapsed marginals, although rare high spread groups and some false positive spread patches remain.

For this fixed ray traced CKM setting, HARP-NET CKM is accurate, fast, and interpretable enough to reduce repeated ray traced map generation, achieving a measured $113.7\times$ speedup over the local MATLAB path. The broader lesson is methodological: diagnose the target distribution, freeze stable structure in auditable priors, and spend neural capacity on the residual structure those priors cannot explain.

REFERENCES

- [1] Y. Zeng and X. Xu, "Toward environment-aware 6G communications via channel knowledge map," *IEEE Wireless Communications*, vol. 28, no. 3, pp. 84–91, 2021.
- [2] Y. Zeng, J. Chen, J. Xu, D. Wu, X. Xu, S. Jin, X. Gao, D. Gesbert, S. Cui, and R. Zhang, "A tutorial on environment-aware communications via channel knowledge map for 6G," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 3, pp. 1478–1519, 2024.
- [3] W. Khawaja, I. Guvenc, D. W. Matolak, U.-C. Fiebig, and N. Schneckenberger, "A survey of air-to-ground propagation channel modeling for unmanned aerial vehicles," *IEEE Communications Surveys & Tutorials*, vol. 21, no. 3, pp. 2361–2391, 2019.
- [4] A. Saboor and E. Vinogradov, "Millimeter-wave UAV channel model with height-dependent path loss and shadowing in urban scenarios," arXiv:2511.10763, 2025. [Online]. Available: <https://arxiv.org/abs/2511.10763>
- [5] R. Levie, Çağkan Yapar, G. Kutyniok, and G. Caire, "RadioUNet: Fast radio map estimation with convolutional neural networks," *IEEE Transactions on Wireless Communications*, vol. 20, no. 6, pp. 4001–4015, 2021. [Online]. Available: <https://arxiv.org/abs/1911.09002>
- [6] J.-H. Lee, O. G. Serbetci, D. P. Selvam, and A. F. Molisch, "PMNet: Robust pathloss map prediction via supervised learning," in *Proc. IEEE Global Communications Conference (GLOBECOM)*, 2023. [Online]. Available: <https://arxiv.org/abs/2211.10527>
- [7] J.-H. Lee, J. Lee, S.-H. Lee, and A. F. Molisch, "PMNet: Large-scale channel prediction system for ICASSP 2023 first pathloss radio map prediction challenge," in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–2. [Online]. Available: <https://doi.org/10.1109/ICASSP49357.2023.10095257>

- [8] Çağkan Yapar, F. Jaensch, R. Levie, G. Kutyniok, and G. Caire, “The first pathloss radio map prediction challenge,” in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–2. [Online]. Available: <https://doi.org/10.1109/ICASSP49357.2023.10433928>
- [9] Y. Li, C. Zhang, W. Wang, and Y. Huang, “RMTransformer: Accurate radio map construction and coverage prediction,” in *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*, 2025, pp. 1–5. [Online]. Available: <https://doi.org/10.1109/VTC2025-Spring65109.2025.11174709>
- [10] Z. Yang, F. Liu, Y. Jin, and K. Vandikas, “Group equivariant convolutional networks for pathloss estimation,” arXiv:2511.17841, 2025. [Online]. Available: <https://arxiv.org/abs/2511.17841>
- [11] Z. Zhong, Z. Yu, P. Li, J. Lv, C. L. P. Chen, and M. Chen, “PathFinder: Advancing path loss prediction for single-to-multi-transmitter scenario,” *Pattern Recognition*, vol. 179, p. 113725, 2026. [Online]. Available: <https://doi.org/10.1016/j.patcog.2026.113725>
- [12] Y. Gao, T. Wen, W. Xie, J. Du, Y. Zeng, D. Niyato, and S. Xu, “Effective outdoor pathloss prediction: A multi-layer segmentation approach with weighting map,” *IEEE Transactions on Vehicular Technology*, pp. 1–6, 2026. [Online]. Available: <https://doi.org/10.1109/TVT.2026.3658966>
- [13] A. Saeizadeh, M. Tehrani-Moayyed, D. Villa, J. G. B. Jr., P. Johari, S. Basagni, and T. Melodia, “AIRMap: AI-generated radio maps for wireless digital twins,” arXiv:2511.05522, 2025. [Online]. Available: <https://arxiv.org/abs/2511.05522>
- [14] X. Xu and Y. Zeng, “How much data is needed for channel knowledge map construction?” *IEEE Transactions on Wireless Communications*, vol. 23, no. 10, pp. 13 011–13 021, 2024.
- [15] M. Shahid, K. Das, H. Ushaq, H. Zhang, J. Song, D. Qiao, S. Babu, Y. Guan, Z. Zhu, and A. Ahmad, “ReVeal: A physics-informed neural network for high-fidelity radio environment mapping,” in *2025 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2025, pp. 1–10. [Online]. Available: <https://doi.org/10.1109/DYSPAN64764.2025.11115911>
- [16] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Prentice Hall, 2002. [Online]. Available: <https://www.pearson.com/en-us/subject-catalog/p/wireless-communications-principles-and-practice/P2000000003199>
- [17] A. Goldsmith, *Wireless Communications*. Cambridge University Press, 2005.
- [18] COST Action 231, “Digital mobile radio towards future generation systems: Final report,” Office for Official Publications of the European Communities, Luxembourg, Tech. Rep., 1999. [Online]. Available: <https://op.europa.eu/en/publication-detail/-/publication/f2f42003-4028-4496-af95-beaa38fd475f>
- [19] A. Al-Hourani, S. Kandeepan, and S. Lardner, “Optimal LAP altitude for maximum coverage,” *IEEE Wireless Communications Letters*, vol. 3, no. 6, pp. 569–572, 2014.
- [20] 3GPP, “Study on channel model for frequencies from 0.5 to 100 GHz,” 3rd Generation Partnership Project, Tech. Rep. TR 38.901, 2026, specification archive, accessed 15 May 2026. [Online]. Available: https://www.3gpp.org/ftp/Specs/archive/38_series/38.901/
- [21] C.-C. Chiu, A.-H. Tsai, H.-P. Lin, C.-Y. Lee, and L.-C. Wang, “Channel modeling of air-to-ground signal measurement with two-ray ground-reflection model for UAV communication systems,” in *2021 30th Wireless and Optical Communications Conference (WOCC)*, 2021, pp. 251–256.
- [22] G. Yang, Y. Zhang, Z. He, J. Wen, Z. Ji, and Y. Li, “Machine-learning-based prediction methods for path loss and delay spread in air-to-ground millimetre-wave channels,” *IET Microwaves, Antennas & Propagation*, vol. 13, no. 8, pp. 1113–1121, 2019. [Online]. Available: <https://onlinelibrary.wiley.com/doi/10.1049/iet-map.2018.6187>
- [23] B. Huang, Z. Xin, F. Yang, Y. Zhang, Y. Liu, J. Huang, and J. Bian, “Transformer-based air-to-ground mmwave channel characteristics prediction for 6g uav communications,” *Sensors*, vol. 25, no. 12, p. 3731, 2025. [Online]. Available: <https://www.mdpi.com/1424-8220/25/12/3731>
- [24] H. Mi, B. Ai, R. He, A. Bodi, R. Caromi, J. Wang, J. Senic, C. Gentile, and Y. Miao, “Measurement-based prediction of mmwave channel parameters using deep learning and point cloud,” *IEEE Open Journal of Vehicular Technology*, vol. 5, pp. 1059–1072, 2024. [Online]. Available: <https://doi.org/10.1109/OJVT.2024.3436857>
- [25] X. Wang, K. Guan, D. He, Z. Zhang, H. Zhang, J. Dou, and Z. Zhong, “Super-resolution of wireless channel characteristics: A multitask learning model,” *IEEE Transactions on Antennas and Propagation*, vol. 71, no. 10, pp. 8197–8209, 2023. [Online]. Available: <https://doi.org/10.1109/TAP.2023.3305096>
- [26] A. Kendall and Y. Gal, “What uncertainties do we need in Bayesian deep learning for computer vision?” in *Advances in Neural Information Processing Systems*, 2017. [Online]. Available: <https://arxiv.org/abs/1703.04977>
- [27] T. Saleh, D. Petrov, O. Tirkkonen, and V. Räsänen, “Probabilistic path loss predictors for mmwave networks,” in *Proceedings of the IEEE 93rd Vehicular Technology Conference*, 2021, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/VTC2021-Spring51267.2021.9448967>
- [28] D. G. Marti, J. Palacios, J. O. Lacruz, and J. Widmer, “A mixture density channel model for deep learning-based wireless physical layer design,” in *Proceedings of the 23rd International ACM Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems*. ACM, 2020, pp. 53–62. [Online]. Available: <https://dspace.networks.imdea.org/handle/20.500.12761/886>
- [29] Y. Lee, X. Ma, and A. S. I. D. Lang, “Modeling of time-varying wireless communication channel with fading and shadowing,” in *2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall)*, 2024, pp. 1–6. [Online]. Available: <https://doi.org/10.1109/VTC2024-Fall63153.2024.10757531>
- [30] X. Wang, K. Tao, N. Cheng, Z. Yin, Z. Li, Y. Zhang, and X. Shen, “RadioDiff: An effective generative diffusion model for sampling-free dynamic radio map construction,” *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 2, pp. 738–750, 2025. [Online]. Available: <https://arxiv.org/abs/2408.08593>
- [31] P. Kyösti et al., “WINNER II channel models,” IST-4-027756 WINNER II Deliverable, Tech. Rep., 2007. [Online]. Available: <https://www.cept.org/files/8339/winner2%20-%20final%20report.pdf>
- [32] X. Cai, J. Rodríguez-Piñeiro, X. Yin, N. Wang, B. Ai, G. F. Pedersen, and A. P. Yuste, “An empirical air-to-ground channel model based on passive measurements in LTE,” *IEEE Transactions on Vehicular Technology*, vol. 68, no. 2, pp. 1140–1154, 2019. [Online]. Available: <https://doi.org/10.1109/TVT.2018.2886961>
- [33] T. Izydorczyk, F. M. L. Tavares, G. Berardinelli, M. Bucur, and P. Mogensen, “Angular distribution of cellular signals for UAVs in urban and rural scenarios,” in *Proc. EuCAP*, 2019. [Online]. Available: <https://ieeexplore.ieee.org/document/8739910>
- [34] J. Walfisch and H. L. Bertoni, “A theoretical model of UHF propagation in urban environments,” *IEEE Transactions on Antennas and Propagation*, vol. 36, no. 12, pp. 1788–1796, 1988.
- [35] F. Ikegami, S. Yoshida, T. Takeuchi, and M. Umehira, “Propagation factors controlling mean field strength on urban streets,” *IEEE Transactions on Antennas and Propagation*, vol. 32, no. 8, pp. 822–829, 1984.
- [36] R.-T. Juang, “Explainable deep-learning-based path loss prediction from path profiles in urban environments,” *Applied Sciences*, vol. 11, no. 15, p. 6690, 2021.
- [37] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. [Online]. Available: <https://arxiv.org/abs/1709.07871>
- [38] G. Moreno Garcia, “CKMGenerator: Generator of path loss, delay spread and angular spread,” Public GitHub repository, 2026, snapshot commit 8b6c5fd26c30. [Online]. Available: <https://github.com/unworthyzeus/CKMGenerator>
- [39] G. Moreno Garcia, “Final_Code_TFG: Final training and evaluation code,” Public GitHub repository, 2026, snapshot commit 63d12a3d1a05. [Online]. Available: https://github.com/unworthyzeus/Final_Code_TFG
- [40] G. Moreno Garcia, “TFGAllProgress_Tries_and_Attempts: Development history and experimental attempts,” Public GitHub repository, 2026, snapshot commit 7c13efd51708. [Online]. Available: https://github.com/unworthyzeus/TFGAllProgress_Tries_and_Attempts
- [41] G. Moreno Garcia, “TFG_ActualText_LaTeX: Thesis source,” Public GitHub repository, 2026, snapshot tag final-thesis-2026-05-15. [Online]. Available: https://github.com/unworthyzeus/TFG_ActualText_LaTeX